

Creating Visualizations in Reports using Looker Studio

Welcome to this instructional reading on creating visualizations in reports using Google's Looker Studio.

This instructional reading is intended to give you the basic knowledge required to help you complete the hands-on labs and final assignment for Looker Studio if you decide to use this product instead of IBM Cognos Analytics

In this reading, we will discuss:

- Configuring page settings
- Adding new pages and navigating between pages
- Blending data from multiple tables
- Adding controls and filters to reports
- Adding different types of visualizations to reports

Configuring Page Settings

You can configure the settings of the current page by right-clicking on a blank part of the canvas and choosing **Current page settings**.

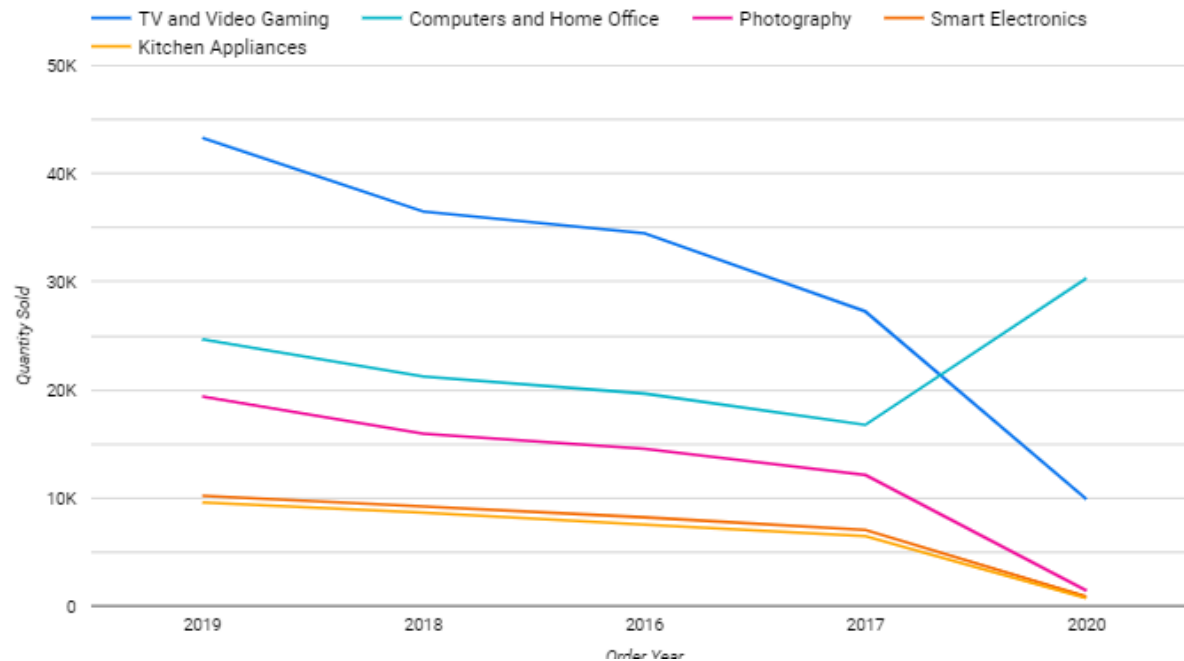
Quantity Sold

396,100

Revenue

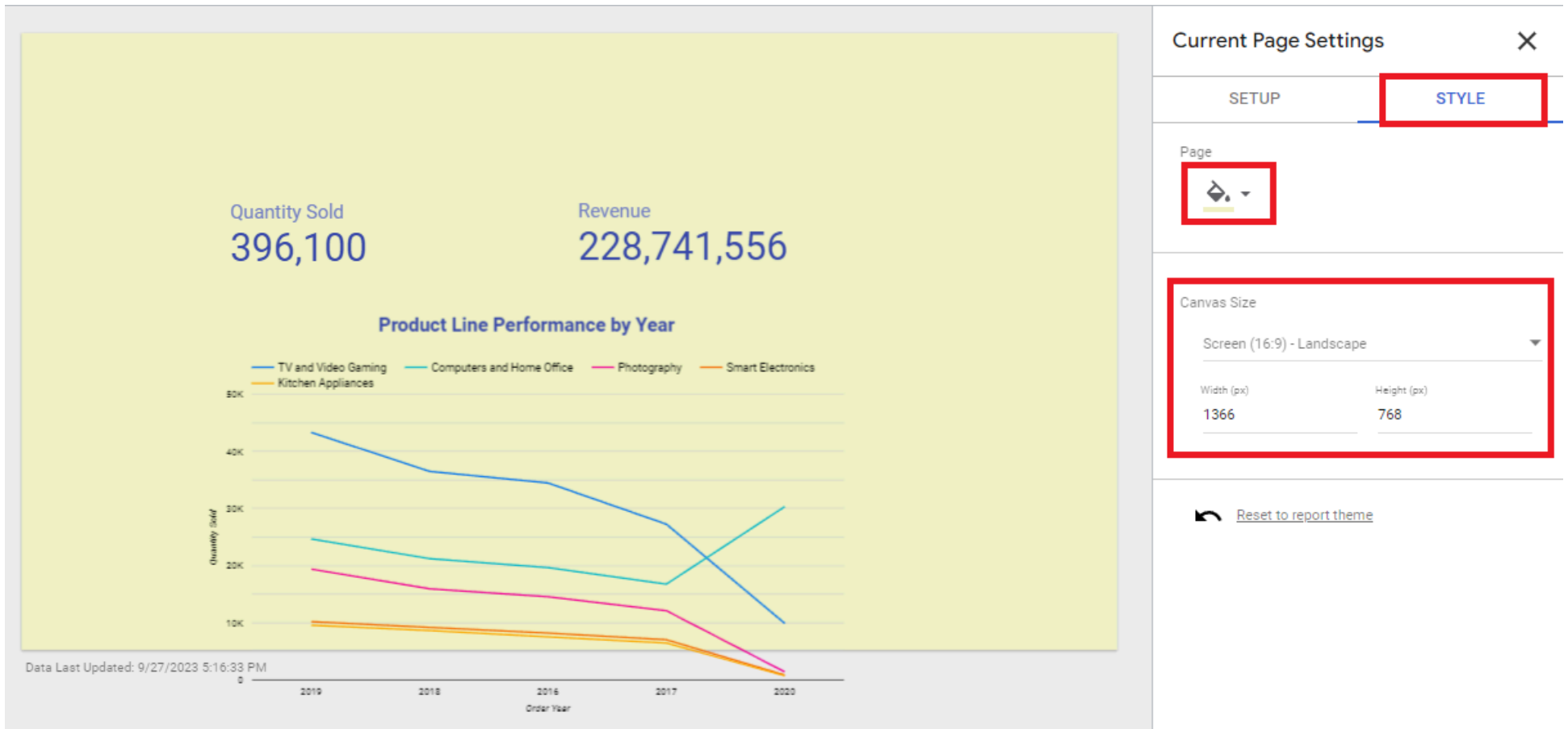
228.7M

Product Line Performance by Year

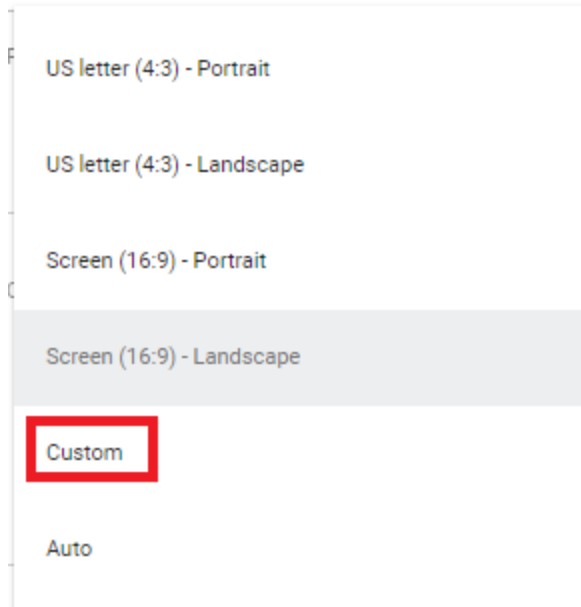


- Select
- Theme and layout
- Report settings
- Current page settings
- Paste
- Download page as PDF

From here, on the **STYLE** tab, you can configure the canvas's background color and dimensions.



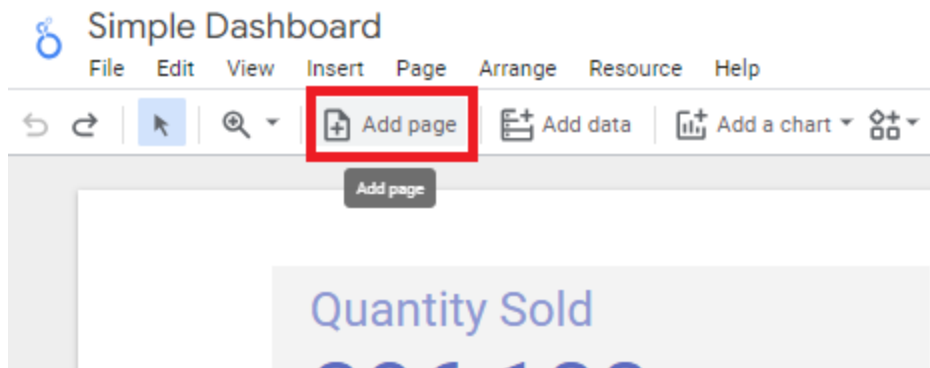
You can choose a ready-made canvas size or create a custom size of your own.



Adding New Pages and Navigating between Pages

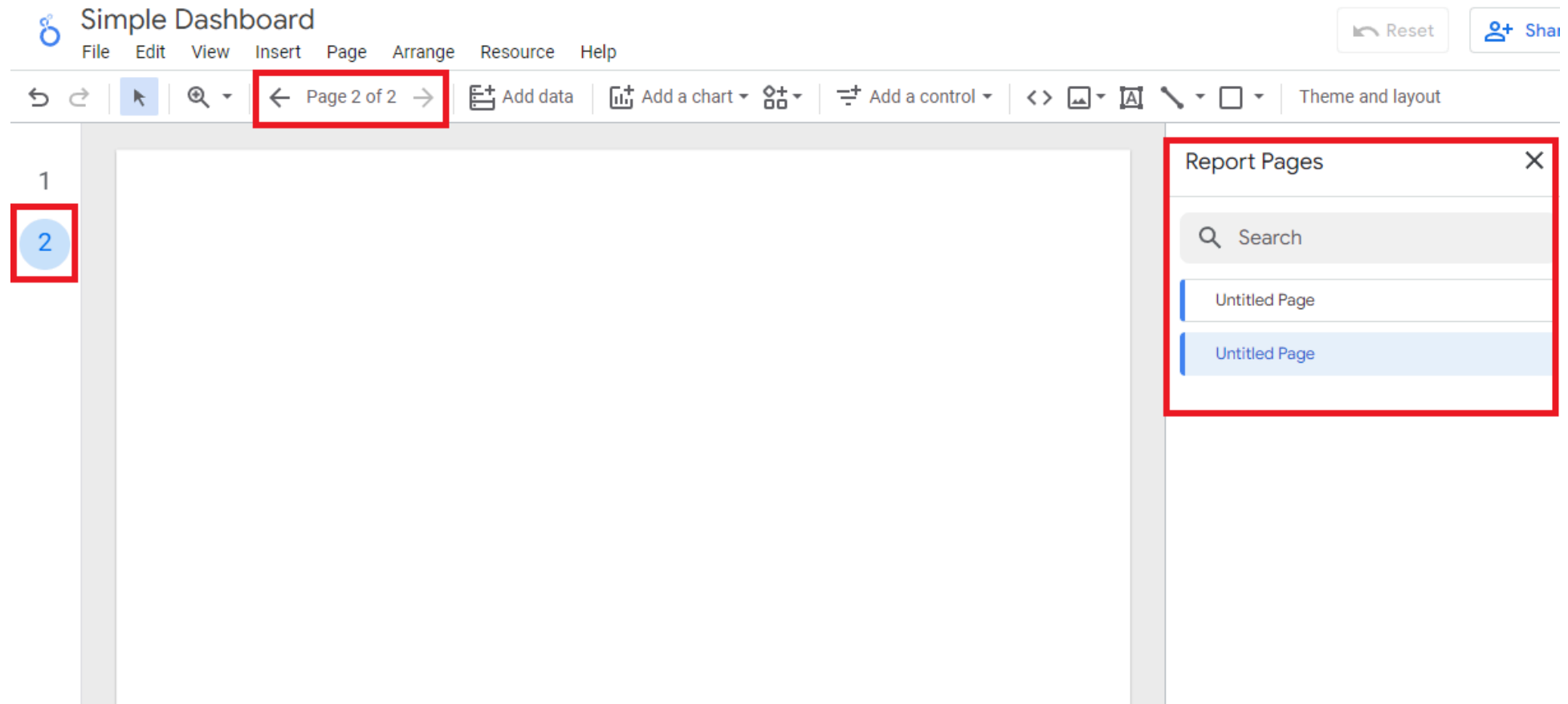
Typically, as you add more visualizations to your reports, you will need more pages in your report to contain them.

To add a new page to your report, either click **Add page** in the toolbar or from the main menu, select **Page**, and then **New page**.

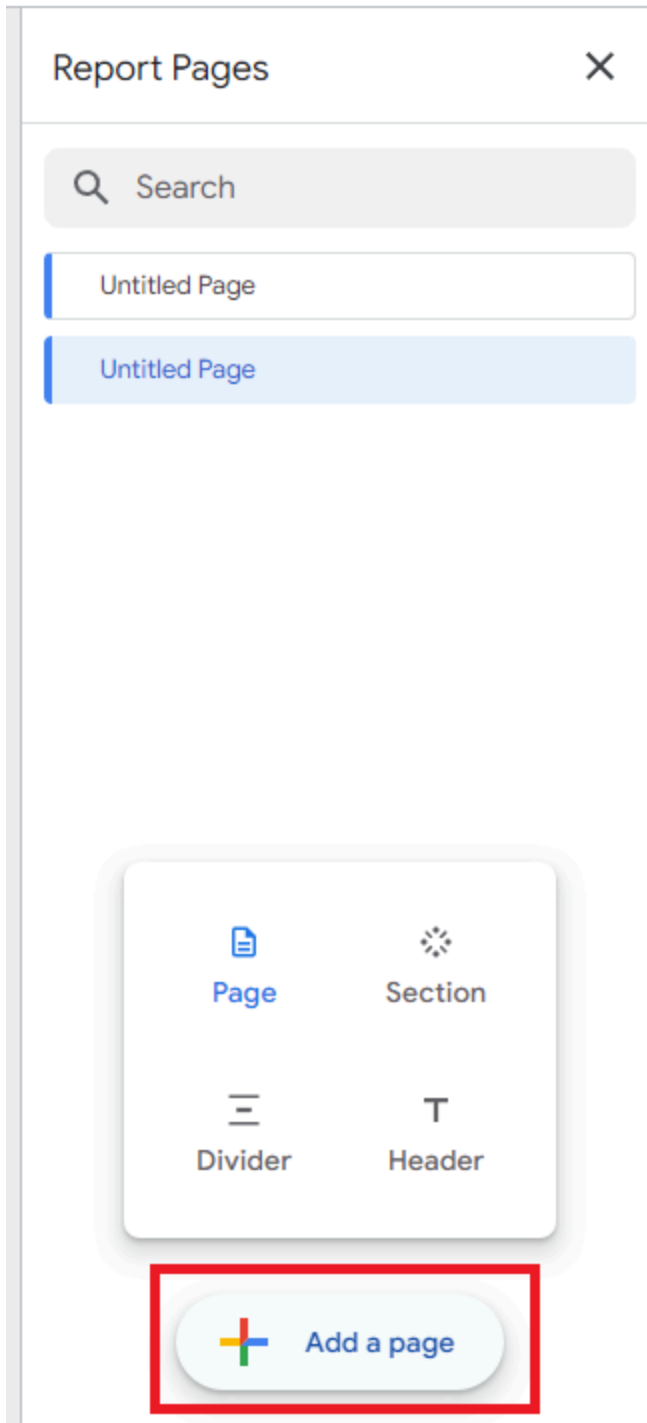


After creating a second page, you will find that the **Add page** item is removed from the toolbar and replaced with a page navigation button.

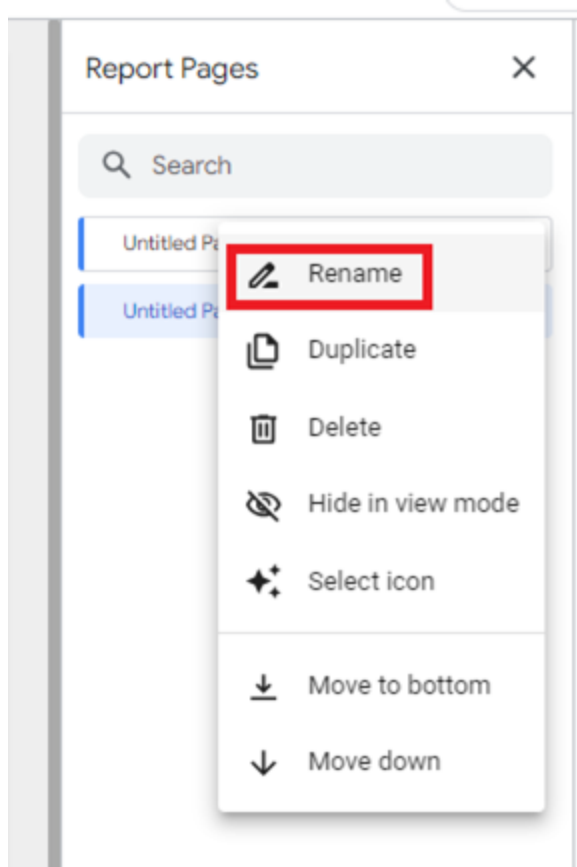
You can use this to navigate between multiple report pages. You can also use the **Report Pages** pane on the right to switch between your pages or the numbered circles on the far left to do the same thing.



To add more pages after the second page, you can now use the menu as before or use the **Add a page** icon below the **Report Pages** pane.



To rename a page, simply click the ellipsis (...) button (the three dots) to the right of an **Untitled Page** for the relevant page in the **Report Pages** pane, select **Rename**, and then enter a new name for the page.

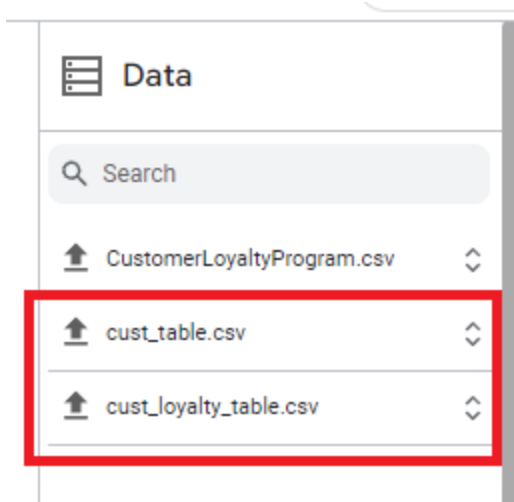


Blending Data from Multiple Tables

There will be occasions where you will have data spread across multiple different tables that you would like to use in a report. Looker Studio allows you to combine these pieces of data together using the **Blend Data** feature.

In Looker Studio, you do this by creating a relationship between the two datasets.

First, you need to add and upload multiple datasets.



Then, having added a chart, you click **BLEND DATA** in the chart's properties pane.

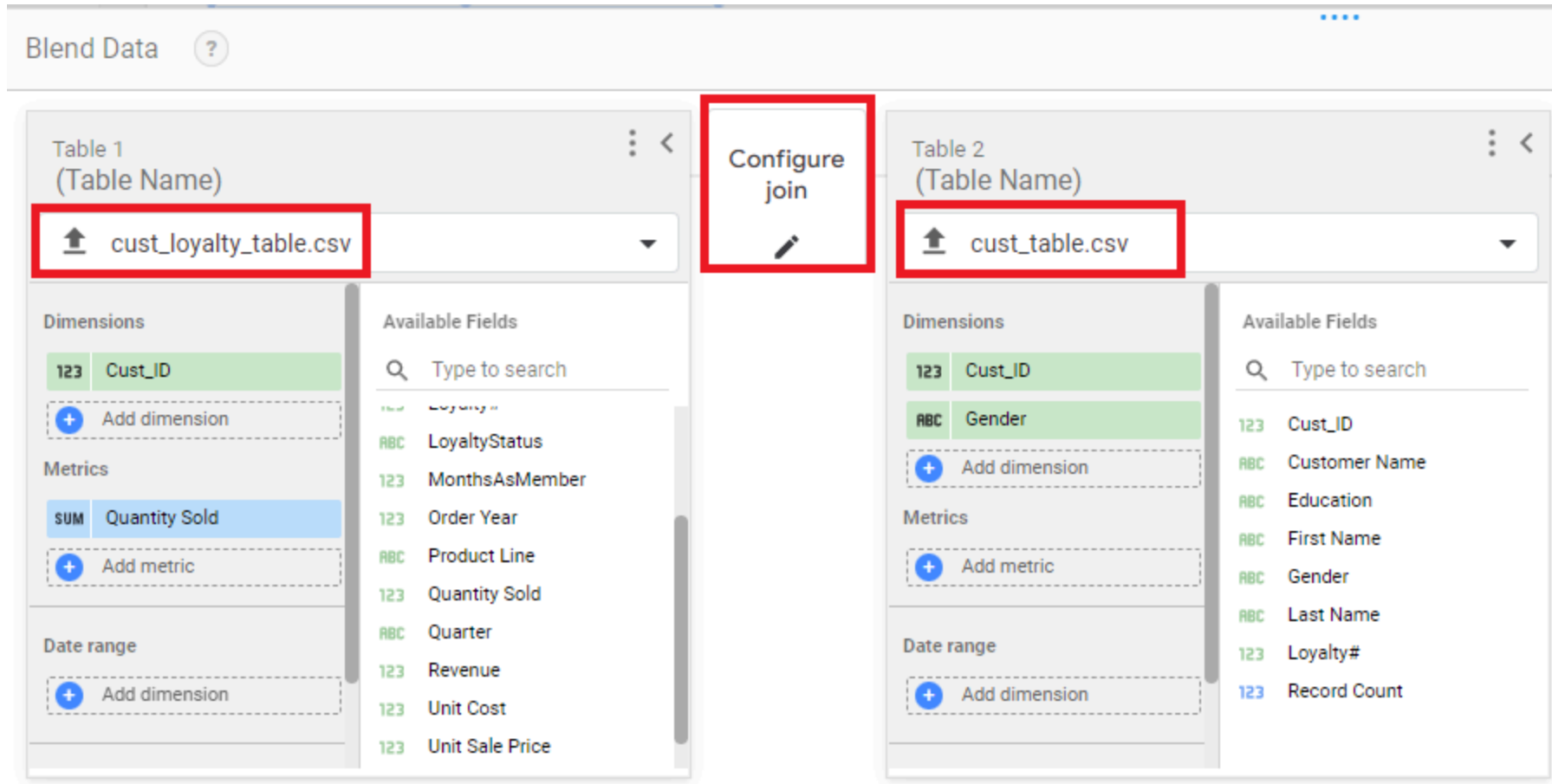
The Simple Dashboard interface shows a bar chart titled "Record Count" with the following data:

Name	Record Count
Paris Norman	50
Sydney Daryl	48
Jamey Loren	48
Leon Casey	45
Toby Dion	45

The right-hand properties pane for the Chart shows the following settings:

- Chart** (selected)
- SETUP** (selected)
- Data source:** cust_table.csv
- BLEND DATA** (highlighted with a red box)
- Date Range Dimension:** Add dimension
- Dimension:** ABC First Name
- Drill down:** (toggle off)
- Breakdown Dimension:** Add dimension

When you select to blend your data, a **Blend Data** window opens at the bottom of the page, which enables you to join another table.



You then configure the join relationship between the two tables by selecting **Configure join**.

Here, you determine how the data rows from the two tables should be joined and how they relate to each other; in other words, which field is common to both of them. When you select to blend your data, a **Blend Data** window opens at the bottom of the page, which enables you to join another table.

Join configuration

Join operator

Tell us how rows from all the tables on the left and the table to the right are combined.

 Left outer	 Right outer	 Inner	 Full outer	 Cross
Returns matching rows from the right table, plus non-matching rows from the left tables				

Join conditions

Tell us how these tables are related. Add one or more fields from the tables to the left that match the fields in the table to the right.

123 Cust_ID (Table 1)	↔	123 Cust_ID (Table 2)
+ Add field	↔	+ Add field

[Cancel](#)[Save](#)

The **Blend Data** window will display the updated join configuration.

Blend Data ?

Table 1
(Table Name)

↑ cust_loyalty_table.csv

Dimensions

123 Cust_ID ↔

+ Add dimension

Metrics

SUM Quantity Sold

+ Add metric

Date range

+ Add dimension

Available Fields

🔍 Type to search

- 123 Loyalty#
- ABC LoyaltyStatus
- 123 MonthsAsMember
- 123 Order Year
- ABC Product Line
- 123 Quantity Sold
- ABC Quarter
- 123 Revenue
- 123 Unit Cost
- 123 Unit Sale Price

1 condition



Table 2
(Table Name)

↑ cust_table.csv

Dimensions

123 Cust_ID ↔

ABC Gender

+ Add dimension

Metrics

+ Add metric

Date range

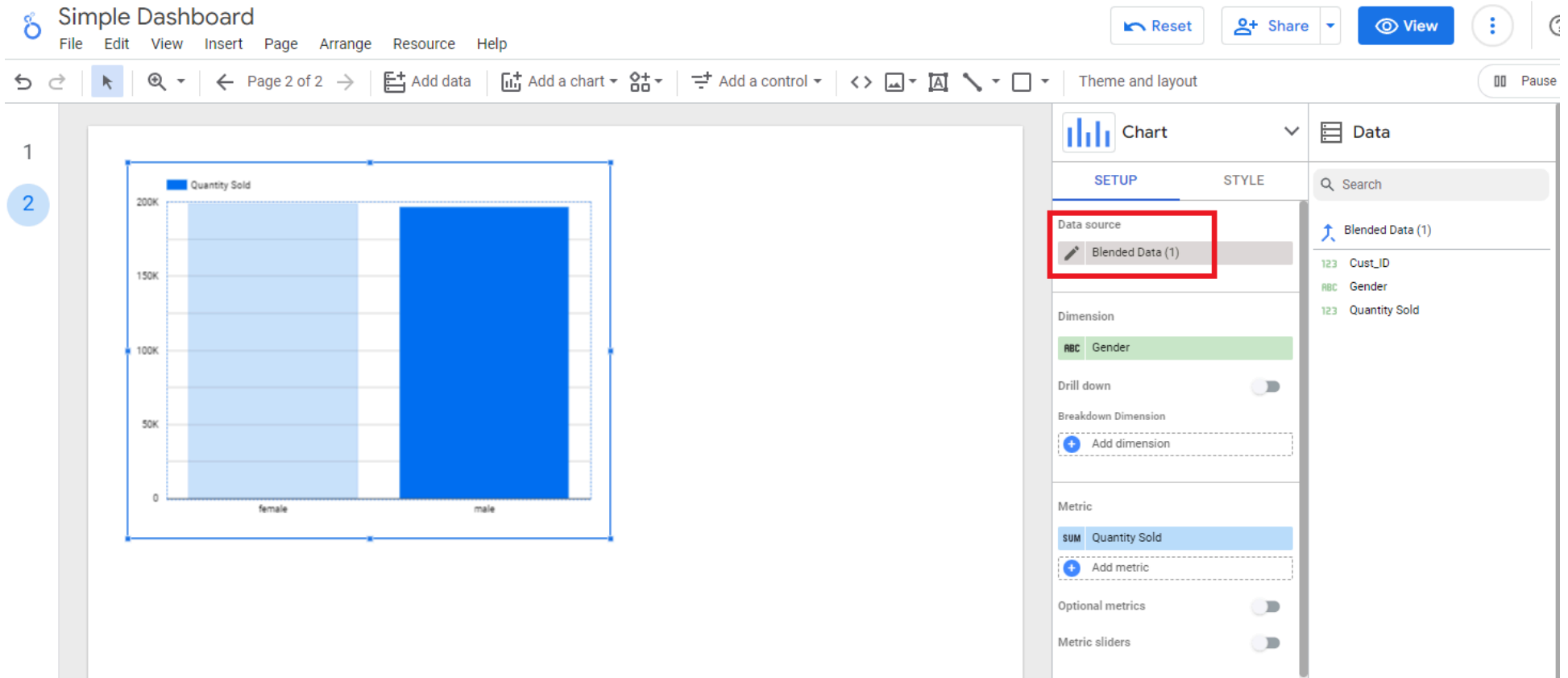
+ Add dimension

Available Fields

🔍 Type to search

- 123 Cust_ID
- ABC Customer Name
- ABC Education
- ABC First Name
- ABC Gender
- ABC Last Name
- 123 Loyalty#
- 123 Record Count

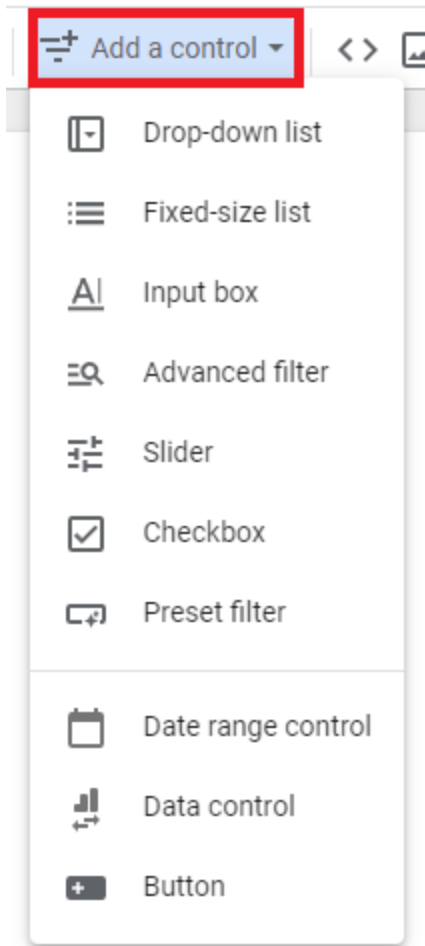
The chart's properties pane will now show **Blended Data** as the data source for your chart, and the chart will now display data that comes from two different data sources.



Adding Controls and Filters to Reports

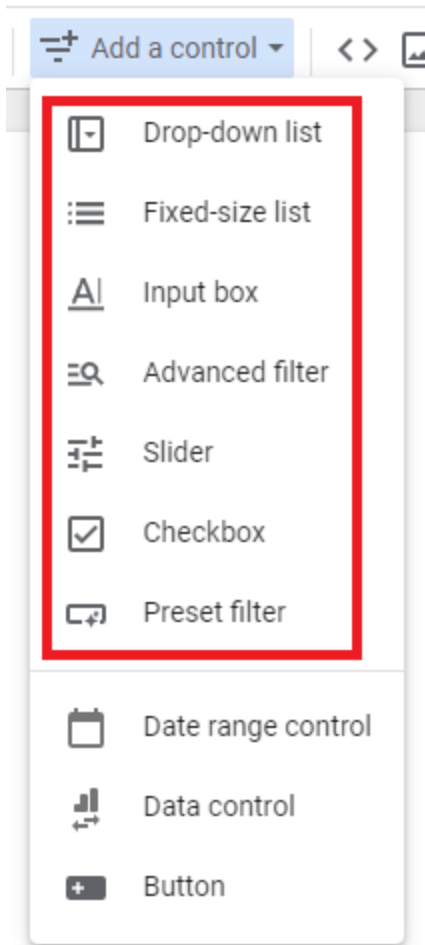
You can add controls and filters to your reports to make them more interactive. Controls can either modify or filter the data displayed in a report or provide a way for you to incorporate user input in a report that allows users to either input data directly into a report or select values or items in the report from a predefined list.

You add controls from the **Add a control** button on the toolbar.

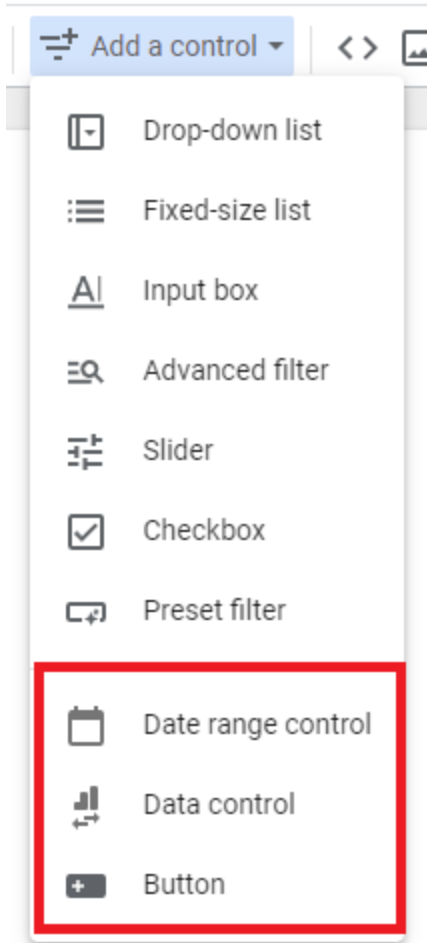


Controls are split into two types.

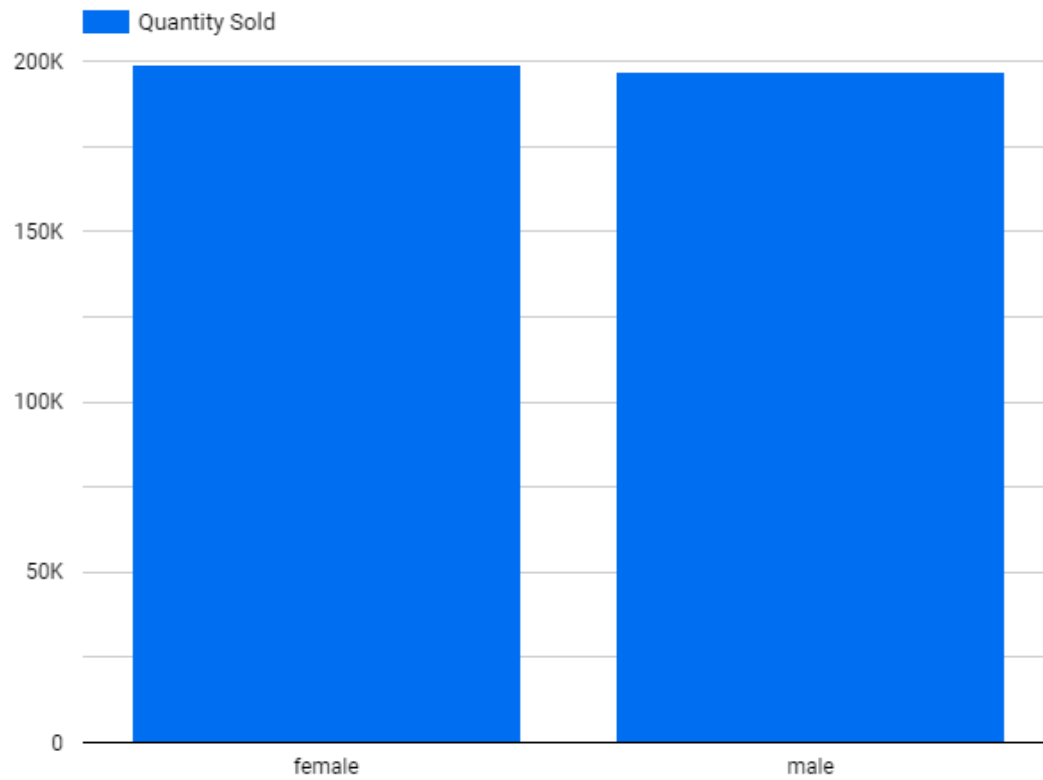
The top subdivision of the menu contains controls that you can use to filter data or configure parameter values, such as drop-down lists, input boxes, sliders, and checkboxes.



The second subdivision of the menu contains controls that you can use to perform specific functions but can't configure parameter values, such as date ranges, data, and buttons.

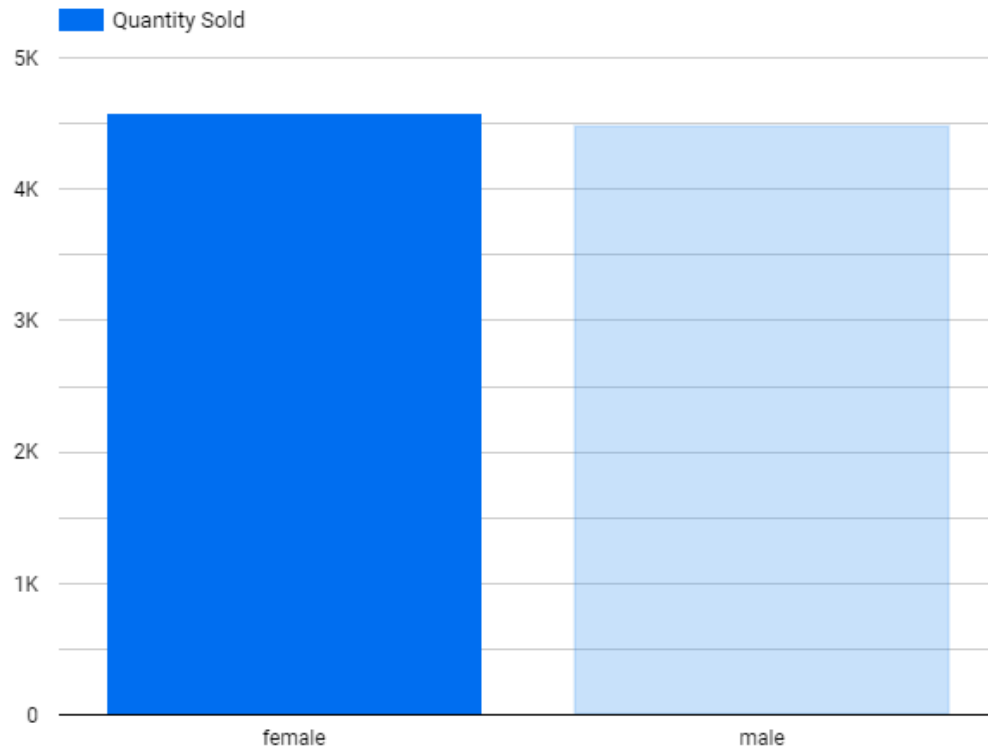


This is an example of a drop-down list control.

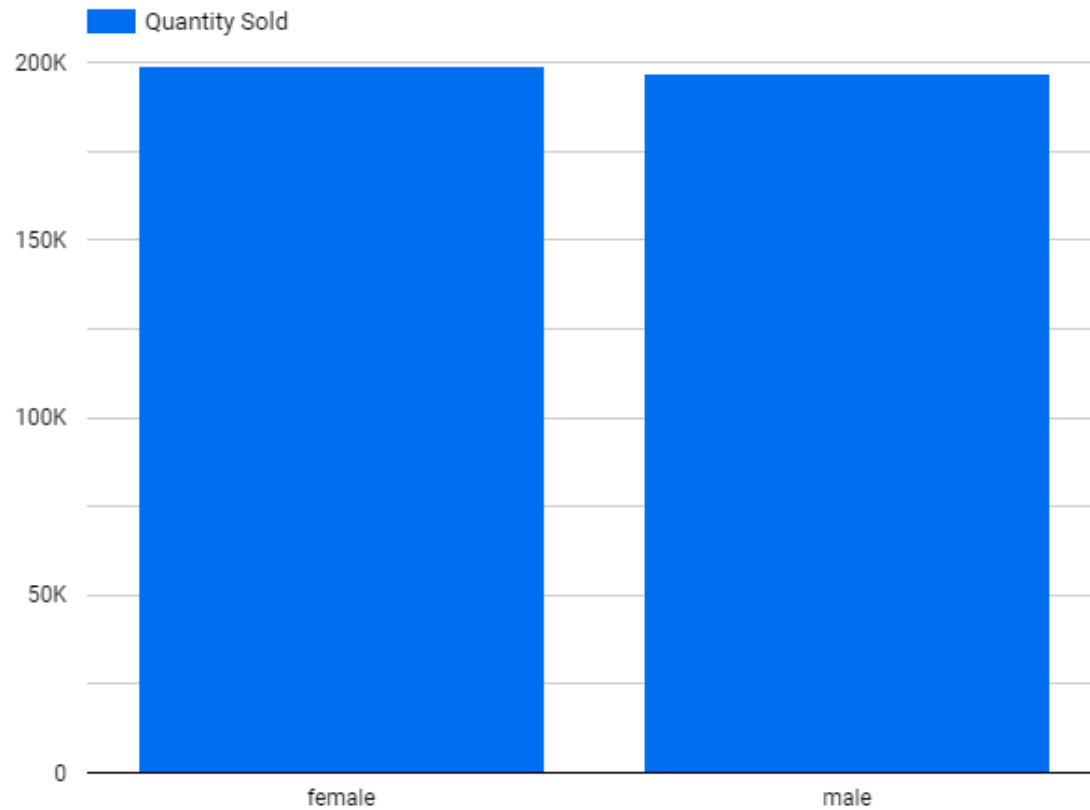


☒ Gender	Quantity Sold
Type to search	
☒ female	199.2K
☒ male	196.9K

This is an example of a slider control.



This is an example of a fixed-size list control.



☒ First Name	Record Count
Type to search	
☒ Paris	51
☒ Norman	49
☒ Sydney	47
☒ Daryl	47
☒ Jamey	47
☒ Loren	45

This is an example of button controls.

Country	Population ▼
China	13,342,720,000
India	11,977,337,695
United States	3,076,528,567
Indonesia	2,388,914,104
Brazil	1,942,727,085
Pakistan	1,715,658,910
Nigeria	1,581,543,983
Bangladesh	1,505,724,822
Russia	1,431,239,304
1 - 100 / 215 < >	

United States



Japan

United Kingdom

Reset filters

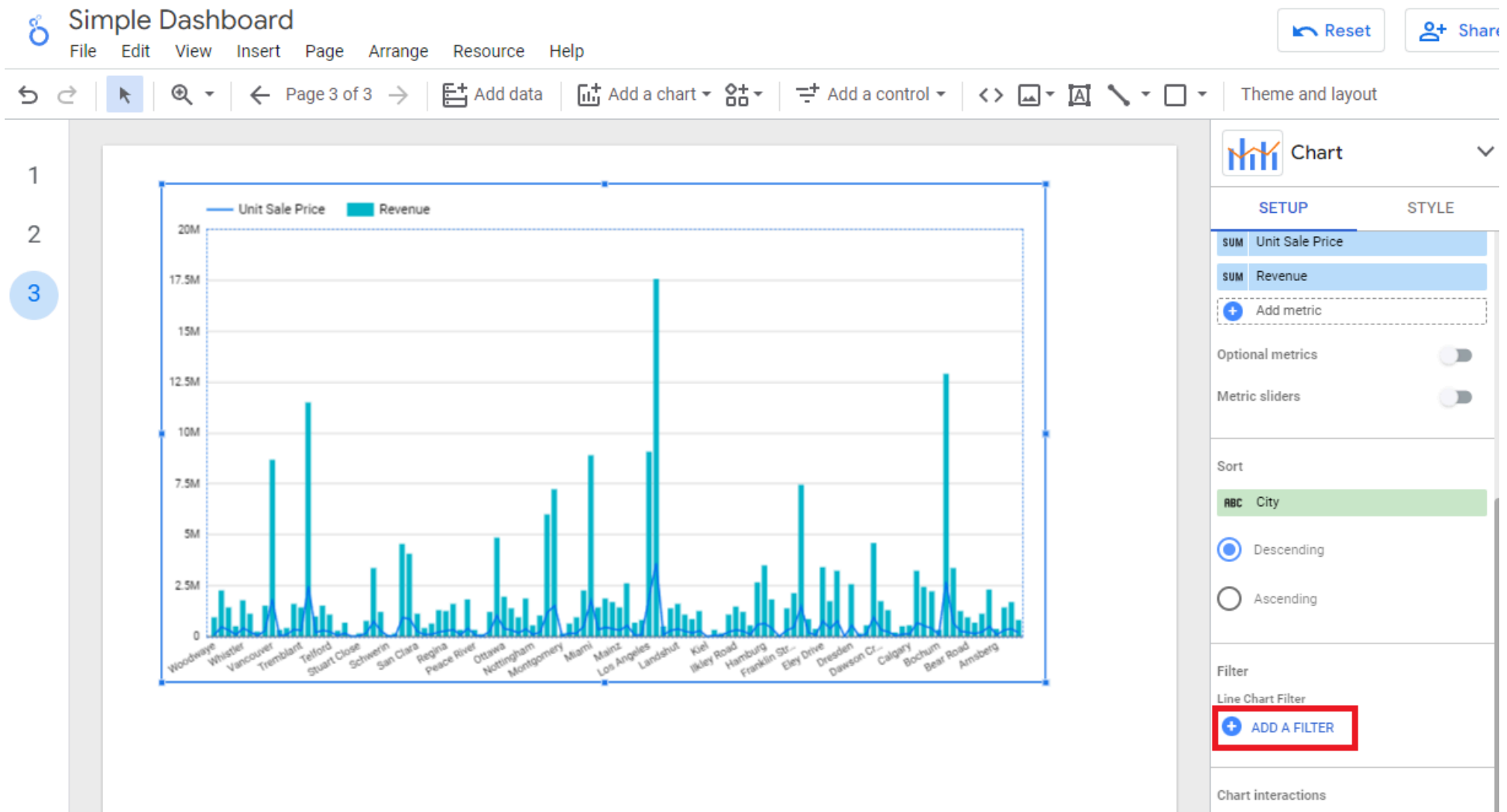
Download report

Share report

View main report

You can add filters to your report to allow you to control what data is or isn't displayed in a report. These can be helpful when you want to temporarily hide some data to focus on other, more important data or just to focus solely on one aspect of the data.

Filters are added directly in the chart's properties pane by scrolling down the pane to the **Filter** section and clicking **ADD A FILTER**.



In the **Create Filter** window, you build your filter by specifying to **Include** or **Exclude**, and then you select the field, the condition, and finally, the value.

You can add further **OR** or **AND** clauses to your filters.

So, in this example, we would exclude any data from the city of Vancouver in our chart.

Create Filter

×

CLOSE

Name

CustomerLoyaltyProgram.csv

Exclude

ABC City

Equal to (=)

Vancouver

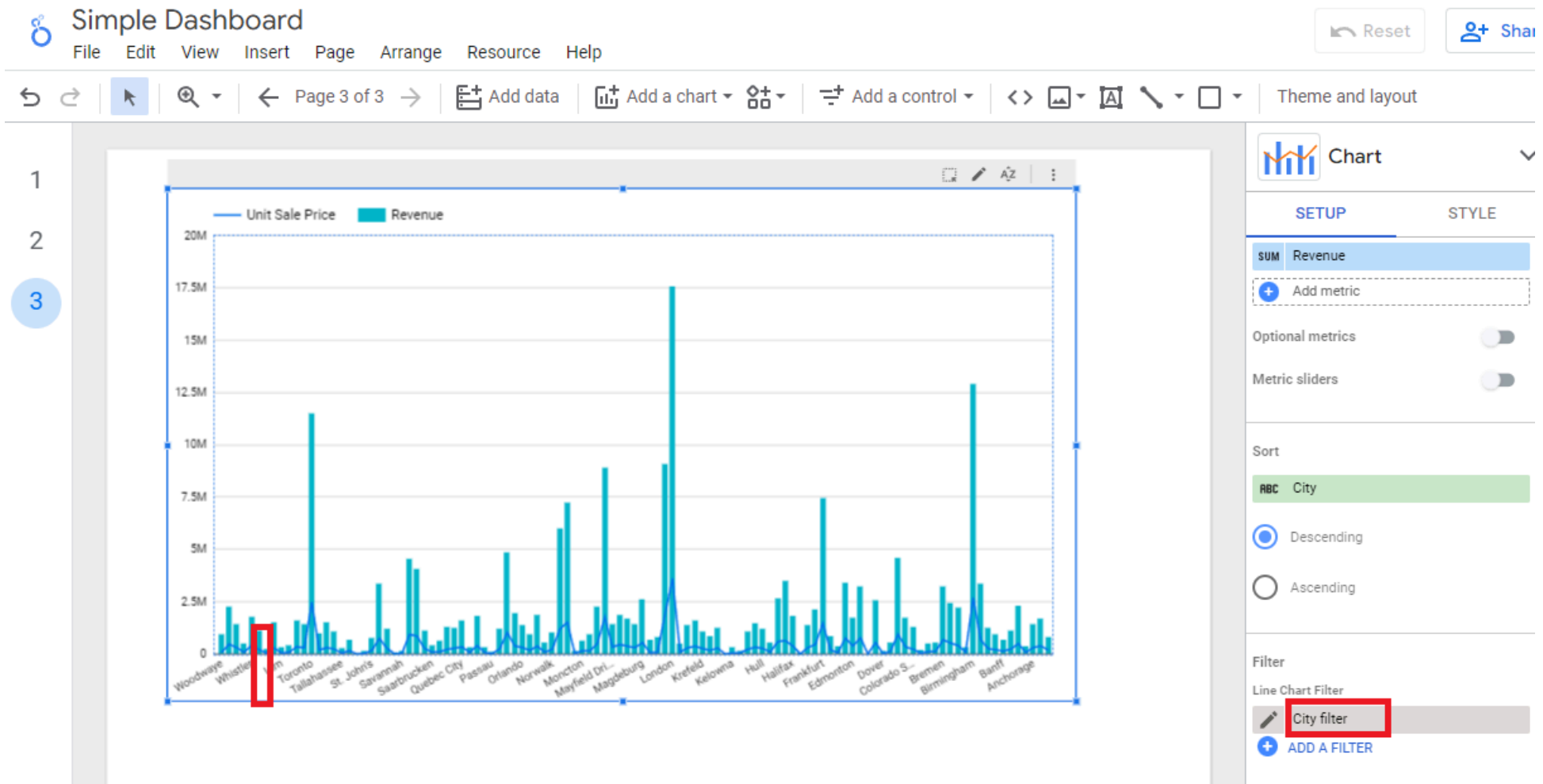
OR

AND

This filter has 1 clause

SAVE

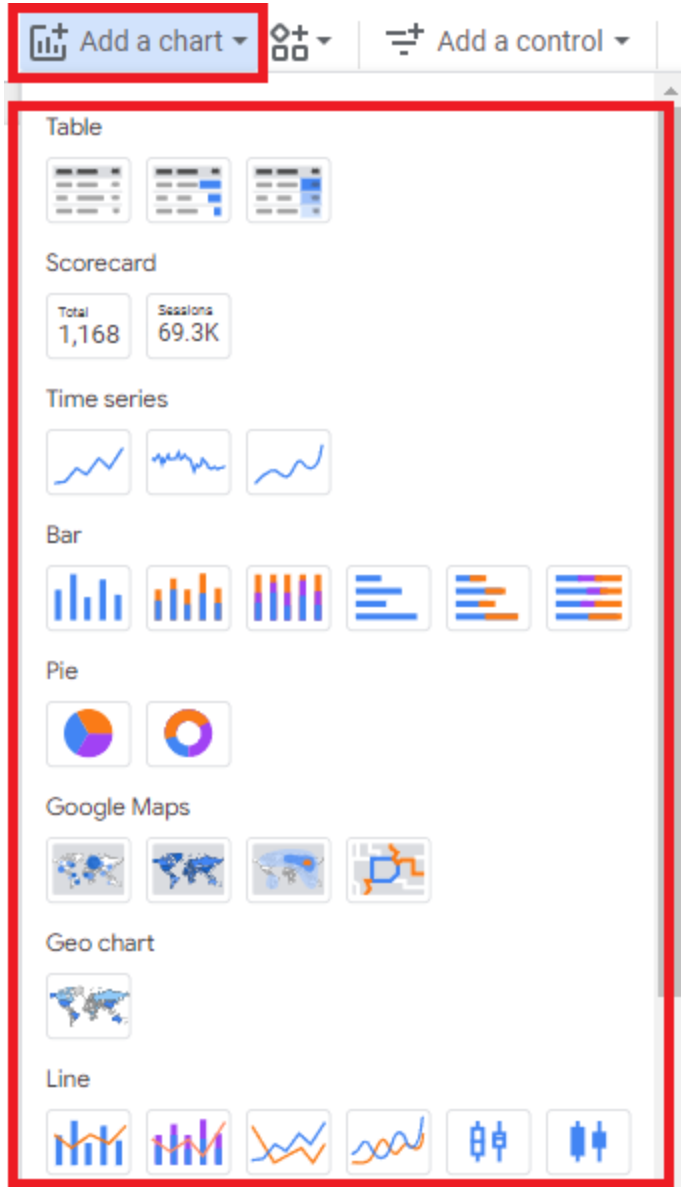
The chart now suppresses the data from Vancouver.



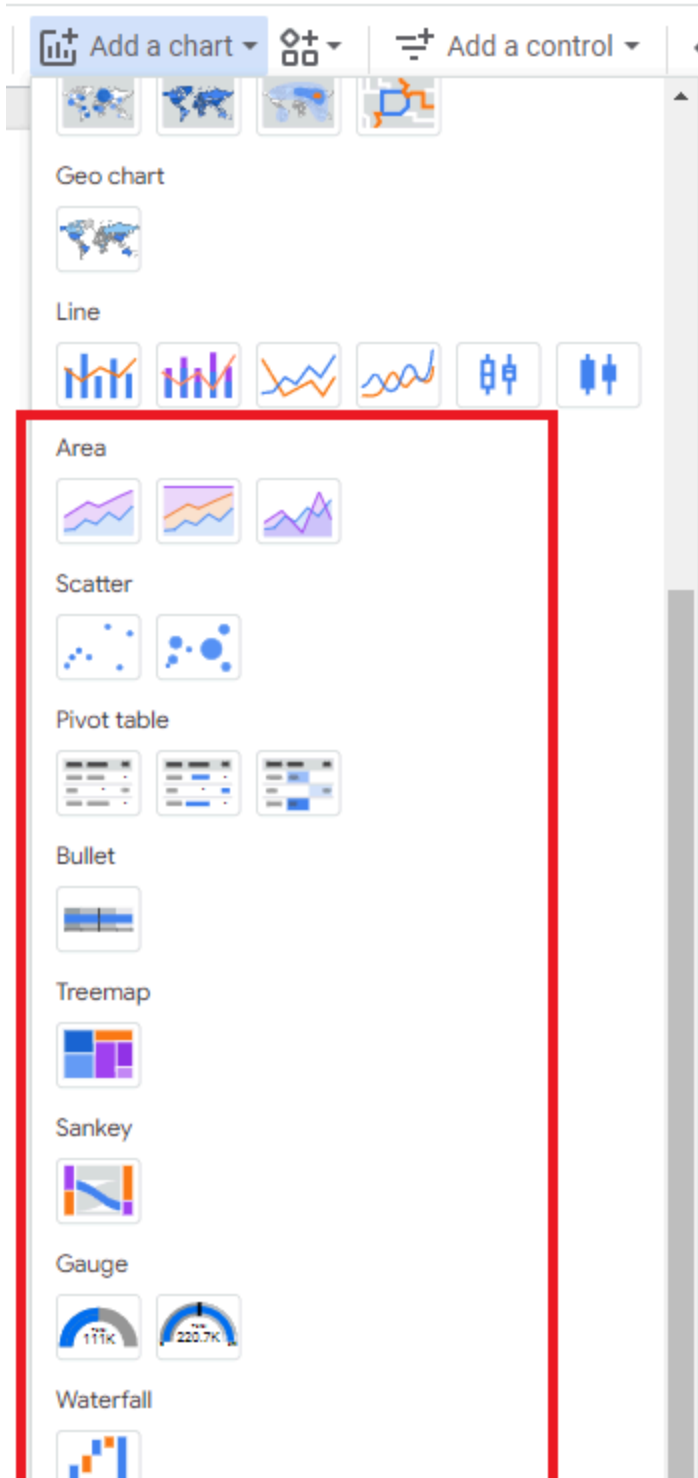
Adding Different Types of Visualizations to Reports

There are many different types of visualizations you can add to your reports.

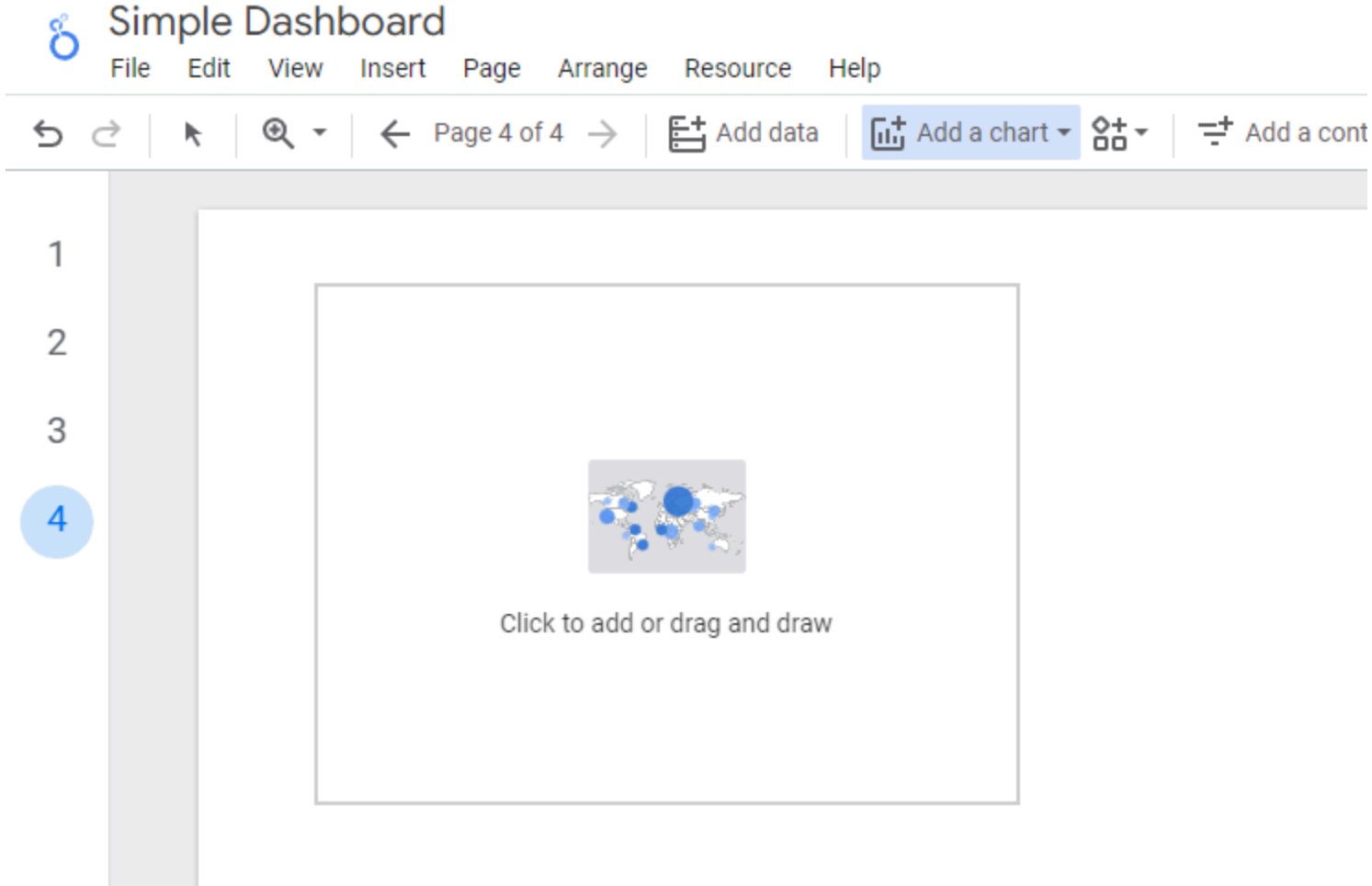
When you click **Add a chart** in the toolbar, you see many different options available, including some of the more common ones, such as tables, scorecards, bar charts, pie charts, line charts, and maps.



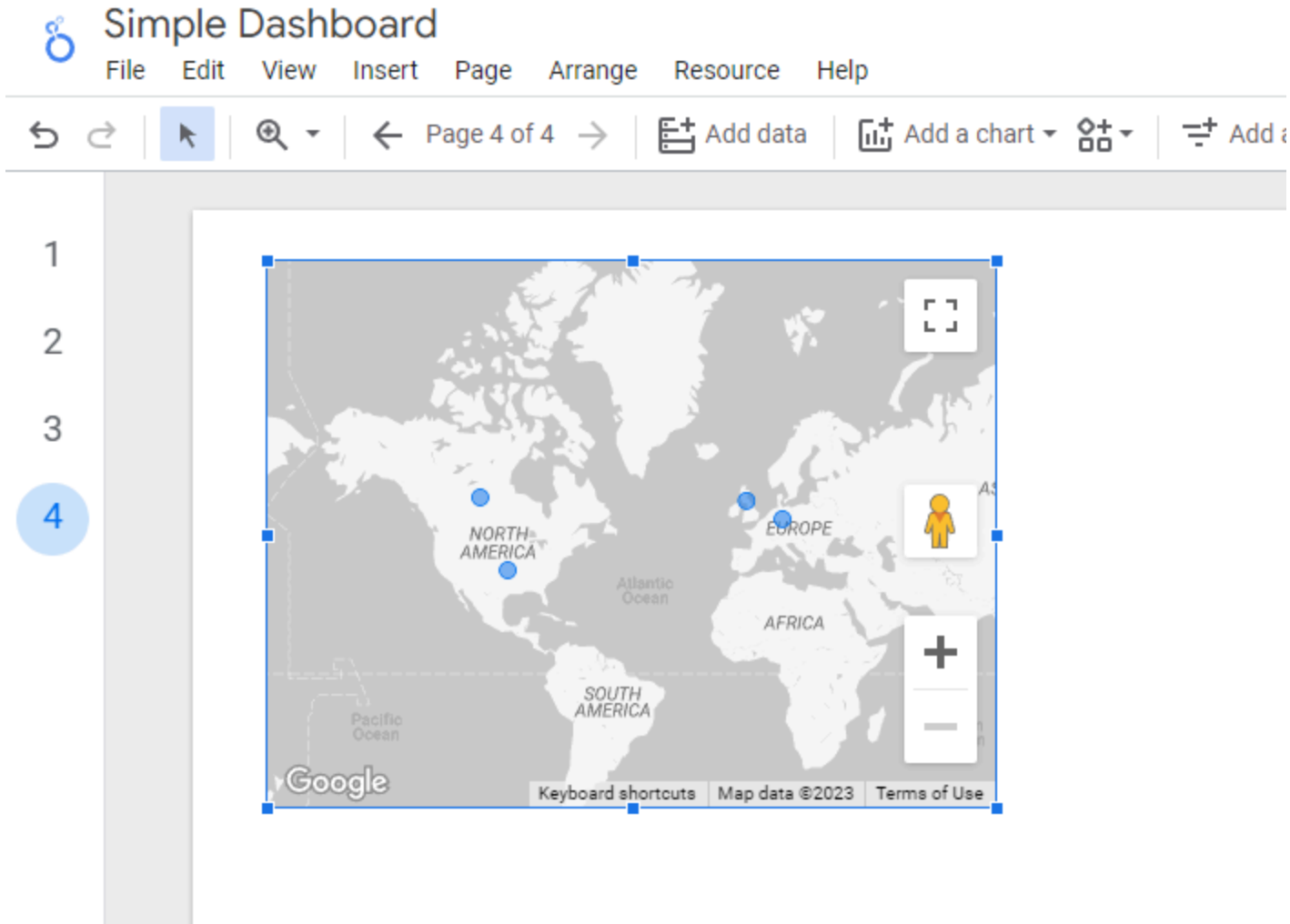
But there are also several other less common types of charts, such as area charts, pivot tables, treemaps, gauges, and waterfall charts.



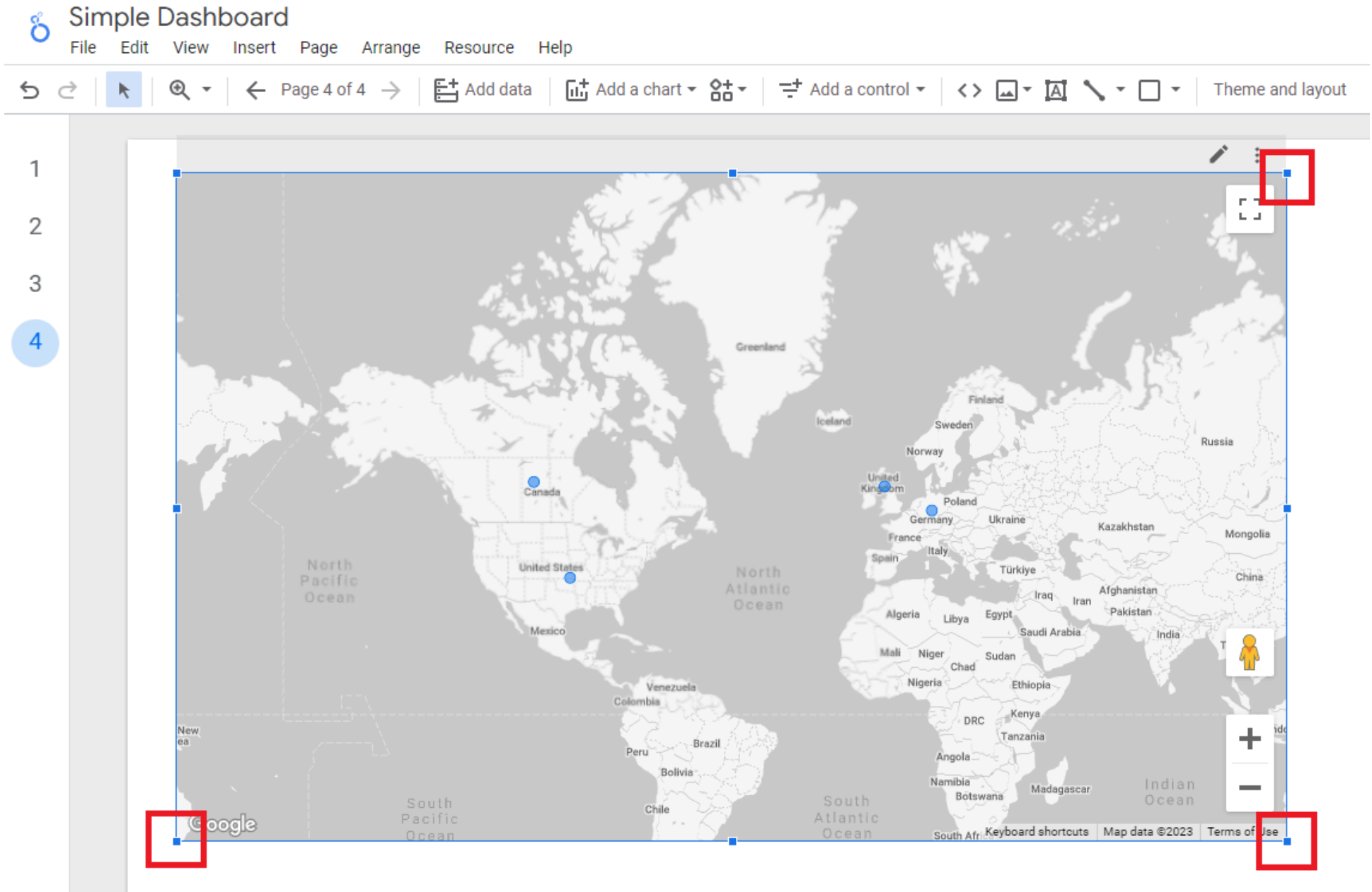
When you select a chart type, the next thing you need to do is place it somewhere on the canvas.



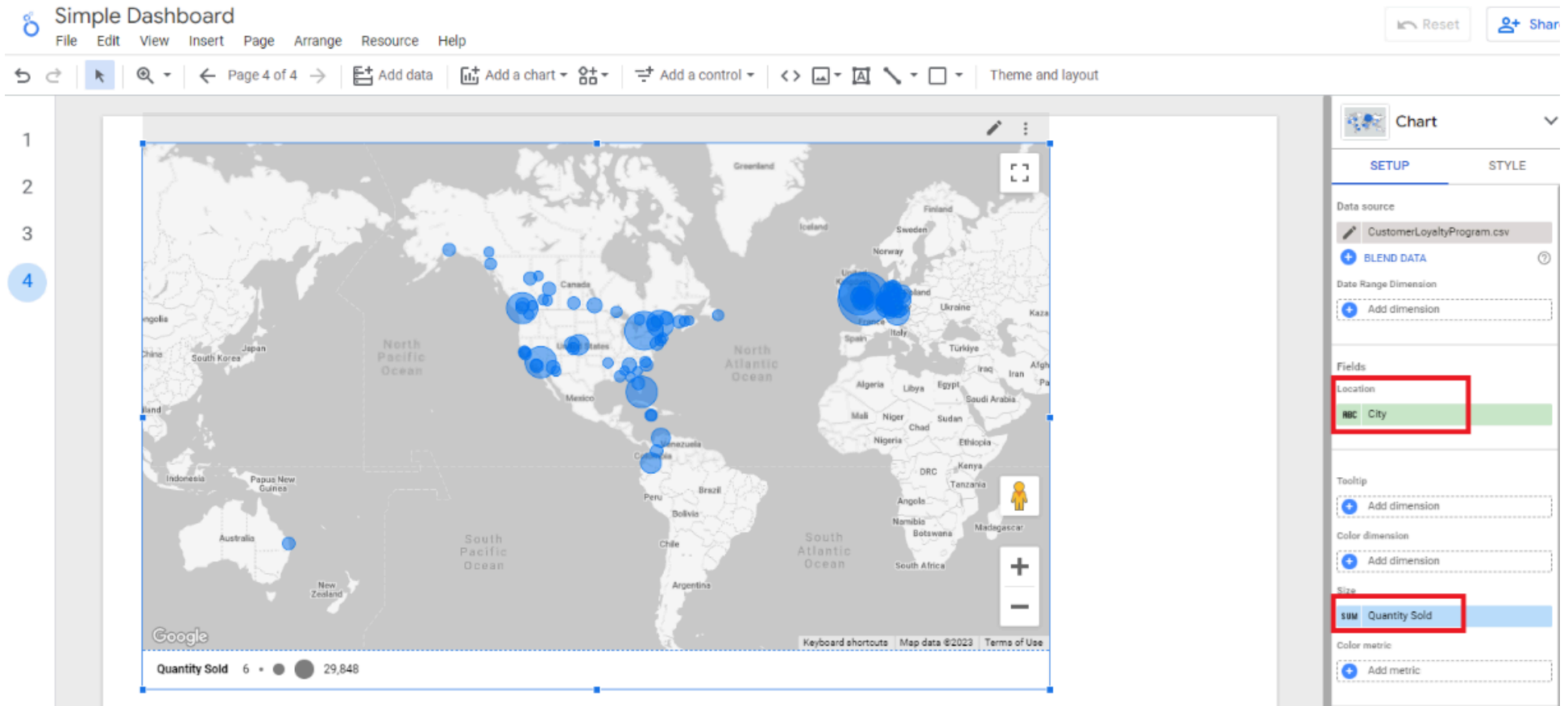
The chart will then display the data from its configured data source.



You can then resize the chart to suit your needs by dragging the handles in the corners.



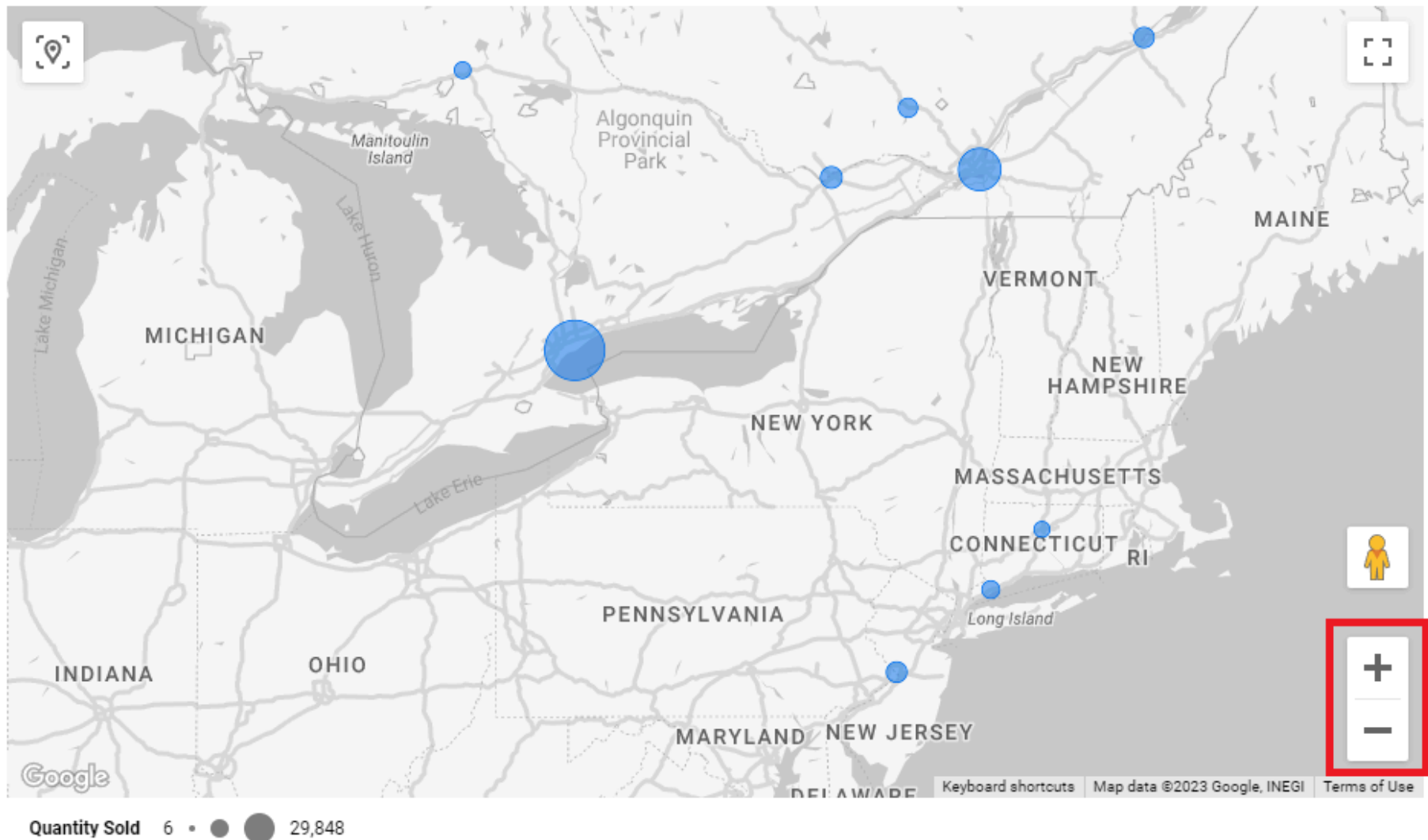
You can then change the configuration of the chart by using the **SETUP** tab of the chart's properties pane to modify which data is referenced by the **Dimension**, **Location**, or **Metric** fields.



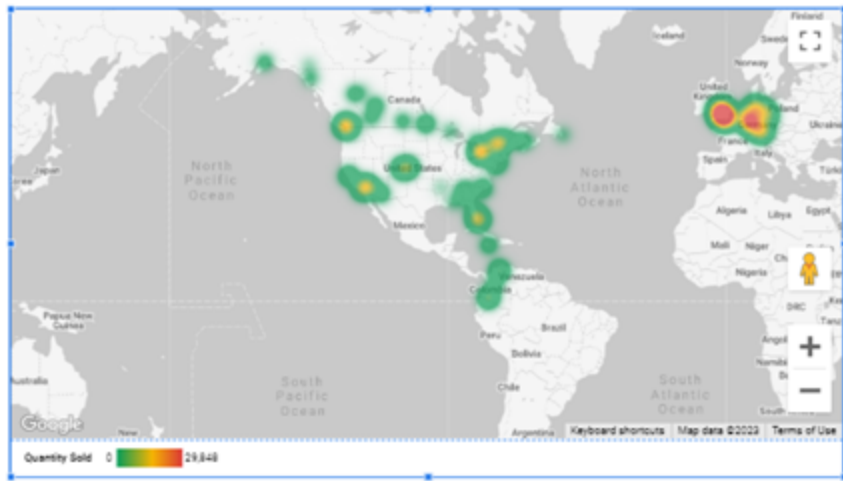
Click the **View** button in the top right of the page to switch to read-only view to get a preview of what the report would look like to an end user.

Click the **Edit** button to switch back to edit mode again.

You can use the mouse wheel (if you have one) or the **+** and **–** symbols on the map to zoom in and out of it in either *View* or *Edit* mode.



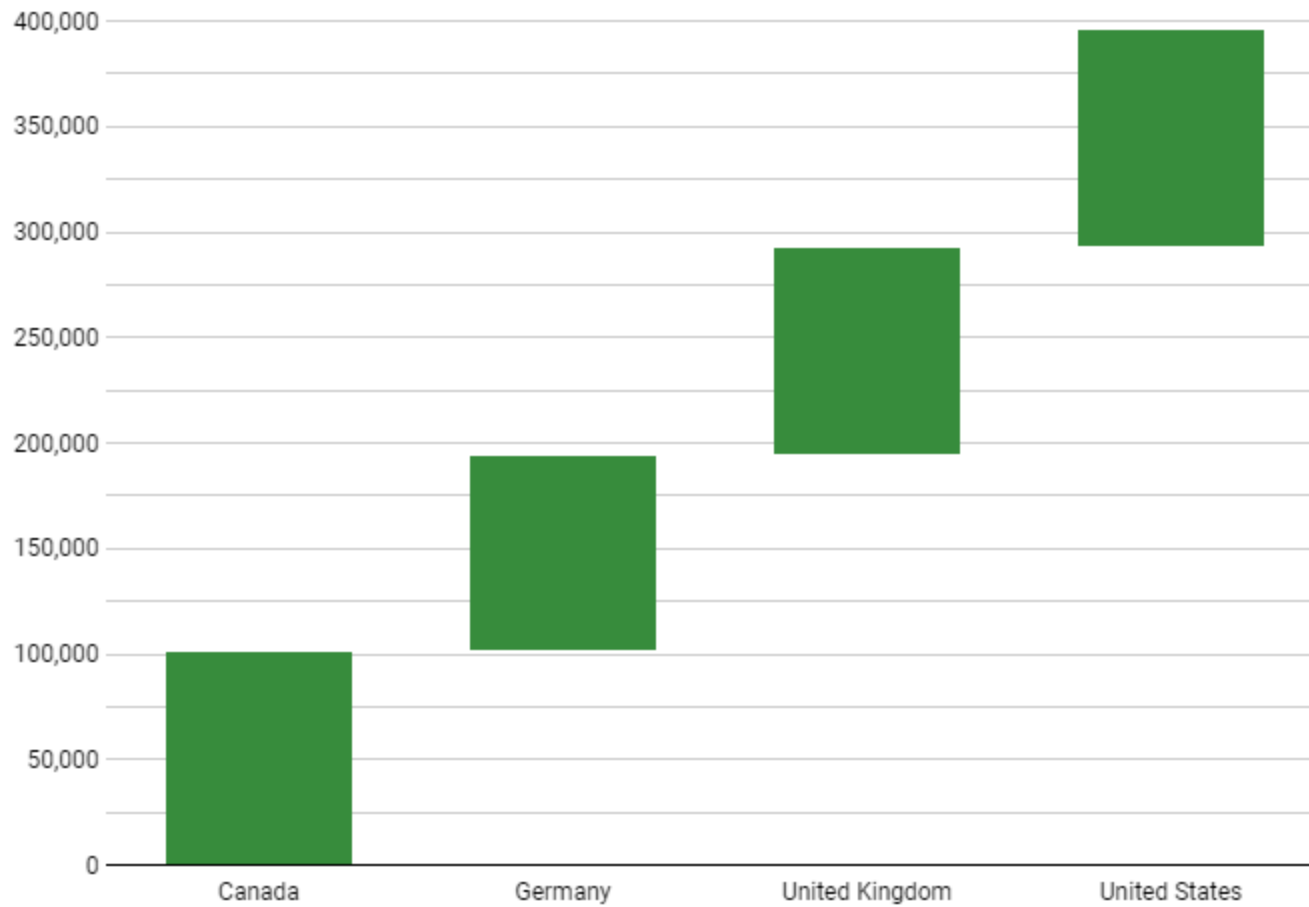
Here is an example of a heatmap chart.



Here is an example of a treemap chart.



Here is an example of a waterfall chart.



✓ Completed